# assignment7 - capacitors

This is a preview of the published version of the quiz

Started: Mar 22 at 7:11pm

## Quiz Instructions

I made a tutorial.

I recommend that first you use the sim for the conceptual questions.

https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab

2 attempts

$$C = \epsilon_0 \frac{A}{d} = \frac{Q}{V}$$

$$E = \frac{\sigma}{\epsilon_0} = \frac{Q}{\epsilon_0 A}$$

$$V = Ed = \frac{Qd}{\epsilon_0 A}$$

$$U = \frac{Q^2}{2C} = \frac{1}{2}CV^2 = \frac{1}{2}QV$$

$$u = \frac{1}{2}\epsilon_0 E^2$$

$$C_{new} = \kappa C_{old}$$

$$E_{new} = \frac{E_{old}}{\kappa}$$

$$\varepsilon = \kappa \epsilon_0$$

$$C_{eq} = C_1 + C_2 + \cdots \quad \text{(parallel)}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \cdots \quad \text{(series)}$$

Question 1 10 pts

For those questions. USe the equations in the header of the assignment. PLay with them

How much charge can be placed on a capacitor of plate area 10 $cm^2$ with air between the plates before it reaches “atmospheric breakdown” where $E = 3.0 \times 10^6$ V/m? ($\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2/N{\cdot}m^2$). hint: easy/

- $5.3 \times 10^{-6}$ C
- $2.7 \times 10^{-8}$ C
- $6.6 \times 10^{-5}$ C
- $4.0 \times 10^{-7}$ C

Question 2 10 pts

The dielectric strength of Rutile is $6.0 \times 10^6$ V/m, which corresponds to the maximum electric field that the dielectric can sustain before breakdown. What is the maximum charge that a $10^{-10}$-F capacitor with a 1.0-mm thickness of Rutile can hold?
hint: easy. there is a video tutorial in the shared folder. in assignment.

- 0.30 mC
- 0.60 $\mu$C
- 6.0 C
- 1.7 nC

Question 3 10 pts

“sandwich” is constructed of two flat pieces of metal (2.00 cm on a side) with a 2.00-mm-thick piece of a dielectric called Rutile ($\kappa = 100$) in between them. What is the capacitance? ($\varepsilon_0 = 8.85 \times 10^{-12}$ $C^2/N{\cdot}m^2$)

- 885 nF
- 8.85 $\mu$F
- 177 pF

100 $\mu$F

Question 4 20 pts

A parallel capacitor with air between the plates has an area A = 2 10^$^{-4}$ m$^2$ and a plate separation d = 1mm. Find its capacitance

1.77 uF

1.77 nF

IDK

`1.77 pF

Question 5 10 pts

**https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab** (https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab) (https://phet.colorado.edu/sims/cheerpj/capacitor-lab/latest/capacitor-lab.html?simulation=capacitor-lab)

Go to simulation and select dielectrics.

Check all the boxes. Increase the voltage to 1.5V

What is true

- energy stays constant, capacitance, energy and charge increase
- capacitance stays constant, energy increases
- capacitance, charge, energy, increase
- electric field stays constant, charge and energy increase

C

D

- A
- B

Question 6 10 pts

leave the voltage at 1.5V

decrease the distance between the plates to 5mm

what is true

- charge, capacitance and energy are multiplied by 2
- charge, capacitance, energy are divided by 2
- charge stays constant but capacitance is multiplied by 2
- capacitance stays constant but energy is multiplied by 2

- C
- D
- A
- B

Question 7 10 pts

Increase the area to 400mm^2. Make sure the dielectrics IS NOT between the plates. move it to the right.

What happens

- energy stays constant and charge is multiplied by 2
- energy stays constant and capacitance is multiplied by 4
- charge, capacitance, energy increases by a factor of 2
- charge, capacitance, energy increases by a factor of 4

- A
- B
- D
- C

Question 8 10 pts

reset all (lower right). check all the boxes. Increase the voltage to 1.5V

Shove the dielectrics between the plates. Observe whats happening

- only the capacitance increases
- charge, energy, capacitance increases and E is 0 inside the dielectrics
- charge and capacitance increase, energy decreases
- charge, energy, capacitance increases and the dielectrics is polarized

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- A
- B
- D
- C

Question 9 10 pts

Building on the previous question. Remove the dielectrics. Disconnect the battery. the capacitor is holding the charge. The charge is constant. they have nowhere to go. E is the electric field between the field

So you have an electric field which is uniform. Now **shove the dielectrics between the plate**s. So we have an insulator placed inside an exterior electric field field. Observe E. What is true

- E is stronger and the dielectrics is not polarized
- E is weaker and the dielectrics is polarized
- E=0 inside the dielectrics
- Nothing happens to E but the dielectrics is polarized

- D
- B
- A
- C

Question 10 10 pts

extra credits

A defibrillator's capacitor is charged to 3000 V and has a capacitance of 32 μF. How much energy is stored in the capacitor?

A) 48 J
B) 96 J
C) 144 J
D) 288 J

Use the equation sheet under capacitance. U is the energy stored inside a capacitor. The capacitance is C.

Unit is farad.

- ○ A
- ○ D
- ○ C
- ○ B

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